Koa Shen

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Intuitive Surgical Mechanical Engineering Hiring Team

Re: Summer 2027 Mechanical Engineering Internship

Dear Intuitive Surgical Hiring Team,

I am a mechanical engineering student at UC Santa Barbara, graduating in June 2028, and I am applying for a Summer 2027 mechanical engineering internship at Intuitive. I am drawn to surgical robotics because it asks engineers to make complex motion precise, dependable, and practical for the people using and maintaining the system. My experience spans mechanism design, kinematic simulation, fabrication, embedded controls, and hands-on validation, and I would be excited to bring that systems perspective to your engineering team.

At SLAC National Accelerator Laboratory, I built a Python/Drake digital twin for a tightly packaged beamline assembly with 30 degrees of freedom and 305 bodies. Millisecond collision checks exposed at least five physical pinch points and informed mechanical redesigns. I developed 15 reusable motion-stage models now being adopted across three assemblies, and redesigned optics mounts, alignment features, and detector strain relief. Pairing simulation with load checks and Solid Edge designs documented with GD&T taught me to connect motion, packaging, and mechanical risk to hardware changes that can be manufactured and inspected.

I have also designed and tested a compact cycloidal actuator: a 15:1 reducer packaged within a NEMA 17 motor's 42 mm square footprint, paired with a TMC2209 driver and motor-shaft encoder. I wrote C++ firmware and bench-validated closed-loop position correction and missed-step detection on a single-joint prototype. With UCSB Gaucho Racing, I led 15 teammates through safety-system design and installation, machined 33 suspension parts to bearing-fit tolerances within 0.0005 inch, and helped reduce a steering-wheel assembly from 5 pounds to 1.7 pounds while integrating driver controls and a display.

These experiences have taught me to work carefully across mechanisms, controls, manufacturing, and test, and to treat clear documentation and serviceability as part of good design. While I have not yet worked on medical devices, I would bring a strong foundation in mechanical engineering, a habit of validating designs against physical behavior, and genuine motivation to learn in a field where reliability matters deeply. I would welcome the opportunity to contribute to Intuitive's surgical robotics work.

Thank you for your consideration.

Sincerely,
Koa Shen